

Exercise 4.2 (Concept: Static Members & Array of Objects)

- Create a package called *library*
- Define two classes in this package: *Book*, *Member*

Book	
- title	: String
- author	: String
- isbn	: String
- publisher	: String
- totalCopies	: int
- availableCopies	: int
- totalBooks	: int
+ Book()	
+ borrowBook()	: void
+ returnBook()	: void
+ displayBookDetails()	: void
+ displayTotalBooks()	: void

Member	
- memberId	: String
- name	: String
- contact	: String
- membershipType	: String
- booksBorrowed	: Book[]
- totalMembers	: int
- maxBorrowLimit	: int
+ Member()	
+ borrowBook(Book b)	: void
+ returnBook(Book b)	: void
+ displayMemberDetails()	: void
+ displayBorrowedBook()	: void
+ setBorrowLimit(int limit)	: void
+ displayTotalMembers()	: void

```

1 package library;
2
3 public class Book {
4     private String title;
5     private String author;
6     private String isbn;
7     private String publisher;
8     private int totalCopies;
9     private int availableCopies;
10    private static int totalBooks;
11
12    public Book(String title,String author,String isbn,String publisher,int totalCopies) {
13        this.title=title;
14        this.author=author;
15        this.isbn=isbn;
16        this.publisher=publisher;
17        this.totalCopies=totalCopies;
18        availableCopies=totalCopies;
19        totalBooks++;
20    }
21
22
23    public boolean borrowBook() {
24        if (availableCopies<=0){
25            return false;
26        }else {
27            availableCopies--;
28            return true;
29        }
30    }
31
32    public void returnBook() {
33        if (availableCopies<totalCopies) {
34            availableCopies++;
35        }else {
36            System.out.println("max copies reached");
37        }
38    }
39    public void displayBookDetails() {
40        System.out.println("bookdetails:"+title+author+isbn+publisher+availableCopies);
41    }
42
43    public static void displayTotalBooks() {
44        System.out.println("totalbooks:"+totalBooks);
45    }
46
47    public String getTitle() {
48        return title;
49    }
50 }
51

```

```

1 package library;
2 import java.util.ArrayList;
3 public class Member {
4     private String memberId;
5     private String name;
6     private String membershipType;
7     private ArrayList<Book> BooksBorrowed;
8     private static int totalMembers;
9     private static int maxBorrowLimit=2;
10    private int borrowCount;
11
12    public Member(String memberId,String name) {
13        this.memberId=memberId;
14        this.name=name;
15        borrowCount=0;
16        BooksBorrowed=new ArrayList<>();
17        totalMembers++;
18    }
19    public void setMaxBorrowLimit(int a) {
20        maxBorrowLimit=a;
21    }
22    public void borrowBook(Book b) {
23        if (borrowCount>=maxBorrowLimit) {
24            System.out.println("limit exceeded");
25        }
26        else {
27            if(b.borrowBook()) {
28                BooksBorrowed.add(b);
29                borrowCount++;
30                System.out.println("book issued");
31            }
32            else {
33                System.out.println("books not available");
34            }
35        }
36    }
37    public void returnBook(Book b) {
38        if(borrowCount<=0) {
39            System.out.println("no books to return");
40        }
41        else {
42            b.returnBook();
43            borrowCount--;
44            for(Book book:BooksBorrowed) {
45                if (book!=null) {
46                    if (book==b) {
47                        BooksBorrowed.remove(b);
48                    }
49                }
50            }
51            else {
52                break;
53            }
54        }
55    }
56 }

```

```

27     if(b.borrowBook()) {
28         BooksBorrowed.add(b);
29         borrowCount++;
30         System.out.println("book issued");
31     }
32     }else {
33         System.out.println("books not available");
34     }
35 }
36 }
37 public void returnBook(Book b) {
38     if(borrowCount<=0) {
39         System.out.println("no books to return");
40     }else {
41         b.returnBook();
42         borrowCount--;
43         for(Book book:BooksBorrowed) {
44             if (book!=null) {
45                 if (book==b) {
46                     BooksBorrowed.remove(b);
47                 }
48             }else {
49                 break;
50             }
51         }
52     }
53 }
54 }
55 public void displayMemberdetails() {
56     System.out.println("memberdetails:"+name+memberId);
57 }
58 public void displayBorrowedBook() {
59     for(Book book:BooksBorrowed) {
60         if (book!=null) {
61             System.out.println("book"+book.getTitle());
62         }
63     }
64 }
65 public void displayTotalMembers() {
66     System.out.println(totalMembers);
67 }
68 }
69 }
70

```

Exercise 4.3 Create another package *librarymanagement*.

Create a Main class with main() function and implement the following:

- i. Import library package.
- ii. Create 2 books (2 copies each) – book1, book2
- iii. Create 2 members
- iv. Set max borrow limit to 2
- v. Member1 borrows 2 books of book1 -
- vi. Member2 try to borrow book1 (No copy available)
- vii. Member2 try to borrow book2
- viii. Member1 try to borrow the second book (Member1 borrow limit reached)
- ix. Member1 return 1 book.
- x. Member2 try to borrow book1
- xi. Display books borrowed by each member.
- xii. Display total titles and the details of all books
- xiii. Display total members and the details of all books.

```
1 package librarymanagement;
2 import library.*;
3 public class LibraryManagementSystem {
4
5     public static void main(String[] args) {
6         // TODO Auto-generated method stub
7         Book b1 = new Book("b1","rohith","1234","pub1",2);
8         Book b2 = new Book("b2","arjun","5678","pub2",2);
9
10        Member m1 = new Member("616","virat");
11        Member m2 = new Member("345","mahi");
12
13        m1.borrowBook(b1);
14        m1.borrowBook(b1);
15        m2.borrowBook(b1);
16        m2.borrowBook(b2);
17        m1.borrowBook(b2);
18        m1.returnBook(b1);
19        m2.borrowBook(b1);
20        m1.displayBorrowedBook();
21        m2.displayBorrowedBook();
22
23    }
24 }
25
26 }
27
```